

COURSE PLAN

Certified in Core Python

Course Code: IT-CORE PYTHON			
Group: IT			
Overall Duration.: 40:00 H			
S.No	Topic	Description	Duration
Python			
1	Python setup Environment and Basics	• Introduction To Python • Install Python and Environment Setup • Python Indentation • Python Program Structure • Taking Input from the user • Python Variables • Formatting Output Python Data Types • Type Casting	120 Mins
2	Practice Program	• Write a program that prints your name, age, and location. • Take two inputs (name, age), print them in formatted output.	60 Mins
3	Python Operators	• Arithmetic operators • Assignment • Logical • Relational • Bitwise • Identity operators • Membership operators	120 Mins
4	Practice Program	• Write a Python program that takes the marks of five subjects as input from the user. • Area of square - a^2; • Area of rectangle - (lb) • Area of circle - πr^2; • The program should then calculate and display the total marks and the average marks obtained	120 Mins
5	Python Control Statements	• if • if else • elif (ladder if) • Nested if	60 Mins
6	Practice Program	• Write a Python program to check whether a number is negative, positive, or zero. • Write a Python program to check whether a number is divisible by both 5 and 11. • Write a Python program to check whether a number is even or odd. • Write a Python program to check whether a year is a leap year or not.	120 Mins
7	Python Loops	• While loop • For loop	120 Mins

- Nested for ,Nested while
- Break, Continue, pass

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8	Practice Program	• Write a Python program to find all factors of a number. • Reverse program using while loop • Python program to calculate the sum of all numbers from 1 to a given number. • Python program to calculate the sum of all the odd and even numbers within the given range.	60 Mins
9	List and Set	• Define list and set • Difference b/w list and set • Insert items: Append items, Extend list • Remove list item: Remove specified index, Clear the Item • Change list item: Change a range of item values • Join two sets • Duplicate set • Intersection_update()	180 Mins
10	Practice Program	• Python program to interchange first and last elements in a list • Python program to swap two elements in a list • Python – Swap elements in String list • Python – Remove items from Set • Python – Check if two lists have at-least one element common	120 Mins
11	Tuple & Dictionary	• Define tuples and Dictionary • Access tuples • add items • Remove item • Loop tuples • join tuples • Dictionary Items • Accessing Items	120 Mins
12	Practice Program	• Store student details in a dictionary, update marks. • Count frequency of each word in a sentence using Tuple	90 Mins
13	Function	• Explain about Functions • Creating a Function, Calling a Function • Arguments: Parameters or Arguments, Number of Arguments, Arbitrary Arguments, Keyword Arguments • Default Parameter Value • Passing a List as an Argument • Lambda and Recursion Function	120 Mins
14	Practice Program	• Write a Python function to find the maximum of three numbers. • Write a Python function to calculate the factorial of a number (a non-negative integer). • Write a Python function to multiply all the numbers in a list	60 Mins
15	Python Modules	• Introduction Modules • Importing built-in modules (math, random, datetime). • User-defined modules, intro to packages.	60 Mins
16	Practice Program	• Write a Python script to display the various Date Time formats - • Current date and time • Current year • Month of year	60 Mins
17	Assessment	• Program to generate random OTP of 6 digits.	60 Mins

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		Create calculator.py module and use it.	
18	Python Oops	- Class & Object - Constructor - Constructor Type : Default Constructor. Parameter Constructor	120 Mins
19	Practice Program	- Write a Python program to create a person class. Include attributes like name, country and date of birth. - Implement a method to determine the person's age.	60 Mins
20	Python Oops	- Inheritance - Inheritance Types - Polymorphism - Types of Polymorphism	120 Mins
21	Python Oops	- Data Abstraction - Encapsulation	90 Mins
22	Practice Program	Bank account class with deposit,withdraw,show account,modify account methods.	60 Mins
23	Python File & Exception Handling	- Working of open() function - Mode Of Files - Read Files, Write Files, Create Files , Delete Files - Define Exception - try, except, finally work flow	60 Mins
24	Tkinter	- Introduction Tkinter - Explain Tkinter package and Function - Create Tkinter window	60 Mins
25	Widgets	- Labels - Widgets (Entry, Text, Checkbutton, Radiobutton). - Layout management, event handling. - MessageBox	120 Mins
26	Practice Program	Student Registration Form GUI.	60 Mins